

← Go Back

Hardware

GIGA Display Shield

Tutorials

Getting Started with the GIGA Display Shield

GIGA Display Shield GFX Guide

GIGA Display Shield LVGL Guide

GIGA Display Shield Draw Images Guide

GIGA Display Shield Touch Interface

GIGA Display Shield Automatic Screen Orientation with IMU

GIGA Display Shield Microphone Guide

GIGA Display Shield Camera Connector Guide

GIGA Display Shield 3D Engine Tutorial

SquareLine Studio With the GIGA Display Shield

Display Camera Feed on GIGA Display Shield with OpenMV

GIGA Display Shield emWin Guide

GIGA Display Shield AppWizard Guide

GIGA Display Shield Display Image From USB

Home / Hardware / GIGA Display Shield / **Getting Started with the GIGA Display Shield**

Getting Started with the GIGA Display Shield

Learn how to set up and use the GIGA Display Shield and get an overview of its features.

Author · Karl Söderby · Last revision · 06/11/2024

The **GIGA Display Shield** is an accessory shield designed for the **GIGA R1 WiFi** board. With it, you can render fast & sophisticated user interfaces on a **800x480** display with **touch support**.

In this guide you will learn how to set your board up with the GIGA R1 WiFi board & become familiar with the available libraries to control it.

Hardware & Software Needed

- Arduino GIGA R1 WiFi
- Arduino GIGA Display Shield
- Arduino IDE

Overview

To use the GIGA Display Shield, there are currently three supported alternatives to draw on the display.

ON THIS PAGE

Hardware & Software Needed

- Overview +
- Board Package & Libraries +
- Hardware Setup
- Camera Connector
- IMU +
- Microphone
- RGB
- Power Consumption

Option 1: GFX Library

The [Arduino_GigaDisplay_GFX](#) library is a layer library for the [Adafruit_GFX](#) graphic core library. This library makes it easy to draw geometrical shapes, printing values, drawing pixels and more.

 To get started with the GFX library, visit the [GIGA Display Shield GFX Guide](#).

Option 2: LVGL

The LVGL framework supports building more advanced UIs with dropdown menus, interactive buttons, scroll functionality using a grid system.

 To get started, visit the [GIGA Display LVGL Guide](#).

Option 3: ArduinoGraphics

[ArduinoGraphics](#) is a simple library for more primitive drawings with a similar API to the GFX library.

 To get started, visit the [GIGA Display Shield Draw Images Guide](#).

Board Package & Libraries

The GIGA Display Shield requires you to install the Board Package for GIGA boards. along with some

The libraries can be installed via the Arduino IDE's library manager, where you will also discover examples.

GIGA Board Package

To use the shield you will need a [GIGA R1 WiFi](#) board. You also need to install the **GIGA Board Package**, which can be done directly in the Arduino IDE, under "Board Manager". The source code for this Board Package can be found in [this GitHub repository](#).



For more details, see [Getting Started with GIGA R1 WiFi](#).

Arduino_H7_Video

The [Arduino_H7_Video](#) manages the video output and integrates third party frameworks such as [LVGL](#). This library is used to configure and initialize the display and to perform basic draw functions.



This library is included with the GIGA Board Package, and does not need to be installed. The documentation for this library is available [here](#).

ArduinoGraphics

The [ArduinoGraphics](#) is required for drawing operations on the screen.



For source code and issues
with the `ArduinoGraphics`
library, please see the [GitHub
repository](#).

Arduino_GigaDisplayTouch

The `Arduino_GigaDisplayTouch` is
required for reading touch points
on the screen, and is needed when
using the LVGL's framework.



For source code and issues
with the
`Arduino_GigaDisplayTouch`
library, please see the [GitHub
repository](#).

Arduino_GigaDisplay_GFX

The `Arduino_GigaDisplay_GFX`
library is a layer library for the
`Adafruit_GFX` graphics core library.

This library is great for drawing
geometrical shapes, printing text &
values, drawing pixels and so on. It
is recommended for beginners.



Learn more about this library
in the [GIGA Display Shield
GFX Guide](#).

Arduino_GigaDisplay

The `Arduino_GigaDisplay` provides
a large set of examples that can be
used with the shield, many of
which are included in the guides
found in the documentation.

This library is also required to use
the built-in RGB, which you can



This library lists out all recommended libraries as dependencies. Installing it through the IDE will also prompt you to install the libraries listed above.

Arduino_BMI270_BMM150 (IMU)

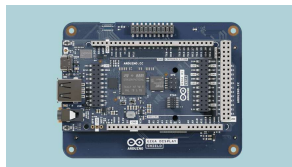
To access the onboard IMU, use the [Arduino_BMI270_BMM150](#) library.

You can read more about its methods in the [library documentation](#).

More information available in the [IMU section](#) in this document.

Hardware Setup

To use the GIGA Display Shield, mount it on the **bottom** side of the GIGA R1 WiFi board. The GIGA R1 WiFi board will be flipped upside down when the display is used.



Bottom View

This makes it possible to freely use the GIGA R1 WiFi's pins while the display shield is connected.

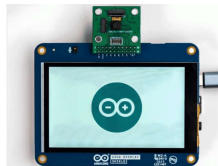


When connecting a module or shield to the GIGA R1 WiFi board, be careful to not connect it at an angle or use

Camera Connector

Located on the top side of the shield is a 20 pin camera connector, which is Arducam® compatible and supports a wide range of cameras. The connector is connected to the [GIGA R1 WiFi's camera connector](#), and is compatible with a wide range of cameras ([see Arducam cameras](#)).

The camera can only be mounted **on the front** of the display shield, meaning it only works in selfie mode. Do not attempt to connect the camera from the back. See the image below for how to connect it:



GIGA R1 + GIGA Display
Shield + Camera
Module.

Read more about this in the [GIGA Display Camera Connector Guide](#), complete with an example.

IMU

This shield has a built-in IMU module, the **BMI270**. This sensor can be used for a number of purposes, for example to automatically orientate an object on the screen, see the following tutorial:

[GIGA Display Shield](#)

To access **BMI270**, use the **BMI270-250** library. Please note that the sensor is not connected to the main I2C bus, so you will need to initialize the sensor on another bus. To do so, simply add the line below to the top of your sketch:

```
1 BoschSensorClass imu(Wir
```

Note that the examples use the default `IMU` (uppercase) class, which you will need to replace.
Example:

```
1 IMU.begin() //for other
2 imu.begin() //for GIGA D
```

Gyroscope

The below example can be used to access the gyroscope data:

```
1  #include "Arduino_BMI2
2  BoschSensorClass imu(V
3
4  void setup() {
5      Serial.begin(9600);
6      while (!Serial);
7      Serial.println("Star
8
9      if (!imu.begin()) {
10         Serial.println("Fa
11         while (1);
12     }
13     Serial.print("Gyrosco
14     Serial.print(imu.gyr
15     Serial.println(" Hz"
16     Serial.println();
17     Serial.println("Gyrc
18     Serial.println("X\t\
19 }
20
21 void loop() {
22     float x, y, z;
23
24     if (imu.gyroscopeAve
25         imu.readGyroscope(
26
27         Serial.print(x);
28         Serial.print('\t')
29         Serial.print(y);
30         Serial.print('\t')
31         Serial.print(z);
32         Serial.println();
33     }
```

Accelerometer

The below example can be used to access the accelerometer data:


```
1 #include "Arduino_BMI2
2 BoschSensorClass imu(V
3
4 void setup() {
5     Serial.begin(9600);
6     while (!Serial);
7     Serial.println("Star
8
9     if (!imu.begin()) {
10         Serial.println("Fa
11         while (1);
12     }
13
14     Serial.print("Accele
15     Serial.print(imu.acc
16     Serial.println(" Hz'
17     Serial.println();
18     Serial.println("Acce
19     Serial.println("X\tY
20 }
21
22 void loop() {
23     float x, y, z;
24
25     if (imu.acceleration
26         imu.readAccelerati
27
28     Serial.print(x);
```

Microphone

This shield has an embedded omnidirectional microphone, **MP34DT06JTR**, which can be used together with the [PDM](#) library. This library is shipped with the GIGA Board Package, so there's no need to manually install it.

Below is a minimal sketch that will print out the samples in the serial plotter (See [Using the Serial Plotter Tool](#) for more information).

```
1  #include <PDM.h>
2
3  // default number of channels
4  static const char channels = 1;
5
6  // default PCM output
7  static const int frequency = 16000;
8
9  // Buffer to read samples
10 short sampleBuffer[512];
11
12 // Number of audio samples read
13 volatile int samplesRead = 0;
14
15 void setup() {
16   Serial.begin(9600);
17   while (!Serial);
18
19   // Configure the data format
20   PDM.onReceive(onPDMReceive);
21
22   // Optionally set the gain
23   // Defaults to 20 or 30 dB
24   // PDM.setGain(30);
25
26   // Initialize PDM with
27   // - one channel (mono)
28   // - a 16 kHz sample rate
```

To get started with the microphone, visit the [GIGA Display Shield Microphone Guide](#).

RGB

This shield has a built in RGB pixel that is controlled via I2C. To use this, install the [Arduino_GigaDisplay](#) library. The following sketch is a minimal example that will blink the blue pixel every second.

```
1  #include <Arduino_GigaD
2
3  GigaDisplayRGB rgb; //c
4
5  void setup() {
6    rgb.begin(); //init t
7  }
8
9  void loop() {
10   rgb.on(0, 0, 255); //
11   delay(1000);
12   rgb.off(); //turn off
13   delay(1000);
14 }
```

Power Consumption

The following test results were recorded with some sample sketches. The setup consist of a GIGA R1 WiFi and a GIGA Display Shield mounted, using the following equipment & software:

[nRF Connect for Desktop](#)

[Power Profiler Kit II](#)

Sketch	Min	Max	Average
GFX	327.20 mA	382.15 mA	363.50 mA
LVGL	294.90 mA	412.49 mA	347.97 mA

Links to the sketches used for these tests are found below:

[GFX Hello World](#)

[LVGL Demo](#)

When selecting an appropriate power source (battery), use the

one of the above sketches with a 1000 mAh battery will only last approx. 2 hours.



Note that there are other factors at play, such as the battery's discharge rate and the general quality of the battery. The above formulas and measurements are to be considered guidelines, please consult the technical information of your battery for more accurate results.

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